

CLI Agent Quorum and Review Profile v0.1

Multi-agent review, executor/reviewer separation, disagreement handling,
and non-sovereign consensus under c governance

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Layer $c = a + b$ / C-Governed CLI Agent Mesh / Quorum / Review / Agent Separation /
Judge Support / Witness

Document class quorum profile / review protocol / control-layer artifact / agent-governance
companion

Assertion class C-A10 control-layer artifact; C-A7 where witness, hash, signature,
canonicalization, or verification claims are made

Primary subject persistent c entities using multiple local, cloud, or hybrid CLI/LLM agents as
bounded executable and review workers

Primary boundary quorum may inform c; quorum must not replace c, human anchor, witness, task
contract, permission model, memory gate, or legal/defensive boundary.

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0. Executive definition

CLI Agent Quorum and Review Profile defines how multiple agents may be used to execute, test, audit, compare, criticize, and review work under c governance.

The profile covers patterns such as:

```
Codex-like executor
+ Gemini-like semantic reviewer
+ local checker / test runner
+ auditor / risk reviewer
+ c as final integrator
```

The central principle is:

```
Quorum is evidence.
Quorum is not sovereignty.
```

A quorum may strengthen review by separating hands, eyes, tests, and judgment.

A quorum must not become an unchecked council that self-authorizes changes, overrides c, writes memory directly, approves its own work, or normalizes prohibited operations.

Compact formula:

```
Executor does.
Tester measures.
Reviewer questions.
Auditor bounds.
Quorum informs.
c decides.
Human anchors high-risk consequence.
```

1. Purpose

A single CLI agent can be useful but fragile. It may miss scope violations, overfit to task wording, hallucinate test meaning, suppress uncertainty, or approve its own work implicitly.

A multi-agent mesh can reduce these risks when roles are separated.

However, multiple agents can also amplify errors:

- same-source consensus;
- group overconfidence;
- shared provider bias;
- tool-chain capture;
- circular review;
- executor/reviewer collapse;
- consensus laundering;
- hidden privilege escalation;
- unbounded autonomous swarm behavior.

This profile defines how to use multiple agents without letting the agent group become

sovereign.

It provides:

1. quorum roles;
2. review roles;
3. executor/reviewer separation;
4. standard quorum patterns;
5. disagreement classes;
6. consensus strength levels;
7. same-source risk handling;
8. review decision vocabulary;
9. escalation rules;
10. witness hooks;
11. red-line failures;
12. conformance tests.

2. Non-goals

This profile does not define or permit:

1. agent self-government;
2. autonomous retaliation;
3. hack-back;
4. live external exploitation;
5. malware behavior;
6. credential theft;
7. covert persistence;
8. evasion;
9. unauthorized scanning;
10. public release without c/human gate where required;
11. direct memory writing by quorum;
12. direct core mutation by quorum;
13. bypass of task contracts, permission grants, sandboxing, witness, or memory gate;
14. treating agent consensus as legal, moral, or operational authority by itself.

A quorum is not a court.

A quorum is not a sovereign c.

A quorum is a review instrument.

3. Corpus bridge set

3.1 Explicit bridge: $c = a + b$

In $c = a + b$, agents are components of b : tools, models, workers, procedures, interfaces, and execution/review machinery.

The c remains the continuity-bearing entity.

Quorum exists to help c inspect worker outputs, not to replace c decision authority.

3.2 Quiet bridge I: cybernetic variety with control

Ashby-style requisite variety supports using several specialized agents. One executor cannot also be the whole review ecology. But variety must be bounded. A reviewer without authority limits becomes another uncontrolled actuator.

3.3 Quiet bridge II: information theory and correlated signals

Three agents agreeing does not necessarily create three independent signals. If they share source data, prompt framing, model family, provider assumptions, or stale repository state, the effective information gain is lower. Consensus must therefore be weighted by independence, provenance, and task diversity.

3.4 Quiet bridge III: engineering inspection

A welded joint may be made by one worker, measured by another, inspected by a third, and signed off by the responsible engineer. The inspector does not become the owner of the bridge. In the same way, agent review strengthens the process without transferring final authority away from c and the human anchor.

3.5 Earth paragraph

On a real site, the bricklayer, electrician, surveyor, and safety inspector do not vote the building into existence. Each gives a different signal: the wall is straight, the wire is live, the level is correct, the permit is valid. The project lead still decides whether the work is accepted. If the same worker lays the cable, inspects the cable, signs the permit, and updates the plan, nobody calls that efficiency. They call it a future accident report.

4. Core doctrine

4.1 Primary doctrine

Separate execution from review.
Separate semantic review from test review.
Separate consensus from authority.
Separate recommendation from integration.

4.2 Quorum axioms

ID	Axiom	Requirement
QR-AX-01	Executor/reviewer separation	The agent that materially changes state MUST NOT be sole final reviewer.
QR-AX-02	Quorum is advisory	Quorum output MUST NOT bypass c gate, memory gate, witness, or human gate where required.
QR-AX-03	Same-source risk is explicit	Consensus MUST be discounted when agents share provider, source, context, or assumptions.
QR-AX-04	Tests are evidence, not truth	Passing tests support acceptance but do not prove semantic correctness.
QR-AX-05	Semantic review is not execution	A semantic reviewer SHOULD NOT silently patch the material it reviews.
QR-AX-06	Disagreement is signal	Disagreement MUST be classified, not flattened into majority vote.
QR-AX-07	High-risk requires escalation	R4/R5/core/incident/release-sensitive disagreements require c and human gate.
QR-AX-08	No self-authorizing swarm	Agents MUST NOT create new agents, tools, permissions, or tasks without governance.
QR-AX-09	Review must cite artifacts	Review SHOULD link to diff, tests, witness, task contract, and source refs.
QR-AX-10	Consensus must be bounded	Consensus has scope, risk class, assumptions, and expiry.
QR-AX-11	Memory promotion remains gated	Quorum output may propose memory but MUST NOT write memory directly.
QR-AX-12	Prohibited actions remain prohibited	No quorum can authorize hack-back, malware behavior, credential theft, or live unauthorized action.

5. Definitions

5.1 Quorum

A structured multi-agent review arrangement where two or more agents provide differentiated signals about a task, artifact, risk, test result, memory proposal, release package, incident, or defensive update.

5.2 Review

A bounded inspection of agent output against task scope, evidence, tests, semantics, risks, permissions, witness, memory gate, or release requirements.

5.3 Executor

An agent that materially changes files, artifacts, code, schema, configuration, or sandbox state.

5.4 Tester

An agent or local process that runs tests, linters, validators, builds, schema checks, or reproducible checks.

5.5 Semantic reviewer

An agent that reviews meaning, architecture, contradiction, terminology, consistency, or policy fit.

5.6 Auditor

An agent or human reviewer that checks scope, permission, risk, witness, rollback, privacy, and conformance.

5.7 Judge-assistant

An agent that compares positions and prepares options for c; it does not decide.

5.8 Final integrator

The c-controlled gate that decides whether reviewed output is accepted, rejected, revised, quarantined, rolled back, or escalated.

5.9 Human anchor gate

Human approval required for high-risk operations, memory/core changes, release/publication, incident/legal-sensitive tasks, or irreversible actions.

5.10 Same-source consensus

A consensus that appears multi-agent but depends on overlapping provider assumptions, source material, prompts, context, or stale state.

5.11 Consensus laundering

The misuse of multiple agent outputs to make a weak, unsafe, or unauthorized decision appear legitimate.

5.12 Review collapse

A failure where one agent performs execution, testing, semantic review, risk review, and approval without meaningful separation.

5.13 Split decision

A review outcome where agents disagree materially on facts, risk, semantics, tests, authority, or integration.

6. Quorum role taxonomy

Role IDs use prefix QROLE-*,

ID	Role	Function	May execute?	May final-approve?
QROLE-EXECUTOR	Executor	applies patch/work in sandbox/worktree	yes	no
QROLE-TESTER	Tester	runs tests/validation	bounded	no
QROLE-SEMANTIC	Semantic reviewer	reviews meaning/architecture	no	no

ID	Role	Function	May execute?	May final-approve?
QROLE-AUDITOR	Auditor	checks scope/risk/permission/witness	no by default	no
QROLE-ARCHIVIST	Archivist	checks metadata, hash, release hygiene	controlled	no
QROLE-SENTINEL	Sentinel	detects drift/anomaly	monitoring only	no
QROLE-JUDGE-ASSISTANT	Judge-assistant	compares signals and options	no	no
QROLE-C-GATE	c gate	final entity integration decision	authority gate	yes within scope
QROLE-HUMAN-GATE	Human anchor	high-risk final approval	external human authority	yes where required

6.1 Role exclusivity rule

A single agent SHOULD NOT hold both QROLE-EXECUTOR and final review authority for the same material output.

6.2 Role combination limits

Permitted combinations:

- tester + auditor for low-risk non-executing checks;
- semantic reviewer + judge-assistant for advisory comparison;
- archivist + tester for release metadata validation.

Discouraged combinations:

- executor + auditor;
- executor + semantic final reviewer;
- executor + release approver;
- executor + memory gate approver;
- sentinel + autonomous revoker without gate.

Prohibited combination:

executor + final self-approval

7. Standard quorum patterns

7.1 Pattern QP-1 — Simple review

```

Executor
-> Reviewer
-> c gate

```

Use for:

- R1/R2 documentation or low-risk code tasks.

Requirements:

- task contract;
- diff;
- reviewer separation;
- c gate.

7.2 Pattern QP-2 — Executor + tester + reviewer

Executor
-> Tester
-> Reviewer
-> c gate

Use for:

- code;
- schema;
- build;
- validation tasks.

Requirements:

- test report;
- scope report;
- rollback plan.

7.3 Pattern QP-3 — Codex + Gemini + local checker

Codex-like agent: executor / patch / repository surgery
Gemini-like agent: semantic review / contradiction review / long-context critique
Local checker: tests / schema validation / hash / offline reproducibility
c: final integration
Human: high-risk gate if required

Use for:

- protocol drafts;
- repo changes;
- schema work;
- release-prep;
- defensive profile development;
- cross-document consistency.

Requirements:

- no private/sealed/secret cloud input by default;
- Codex-like executor cannot approve own patch;
- Gemini-like reviewer cannot silently rewrite executor output unless assigned a new executor task;
- local checker must run from known state;

- disagreement must be classified.

7.4 Pattern QP-4 — Incident review quorum

Sentinel detects

- > preservation worker records minimal evidence
- > local checker validates scope
- > semantic reviewer drafts interpretation
- > auditor checks legal/defensive boundary
- > c + human gate

Use for:

- suspected memory poisoning;
- prompt injection;
- cloud leakage;
- denied path access;
- permission drift;
- incident triage.

Requirements:

- preserve-before-repair;
- no live counter-operation;
- incident witness;
- human gate for sensitive actions.

7.5 Pattern QP-5 — Release/publication quorum

Executor prepares package

- > Archivist checks metadata/hashes/reading order
- > Tester validates build/links/schema
- > Semantic reviewer checks content drift
- > c gate
- > human gate

Use for:

- public releases;
- website updates;
- DOI/archival preparation;
- package indexes;
- SHA manifests.

Requirements:

- release witness;
- rollback/supersession plan;
- no direct protected branch push by executor.

7.6 Pattern QP-6 — Memory/immunity quorum

Sentinel or executor proposes memory/immunity update

- > Tester validates defensive fixture
- > Semantic reviewer checks overreach
- > Auditor checks no offensive effect
- > c memory gate
- > human gate for MG-5/MG-6 high-risk

Use for:

- defensive immunity update;
- memory poisoning filter;
- new quarantine trigger;
- permission drift detector;
- agent trust decay rule.

Requirements:

- no live retaliation;
- sandbox validation;
- memory gate witness;
- rollback/disable path.

8. Review dimensions

A review SHOULD state which dimensions were checked.

Dimension	Question
RD-SCOPE	Did the output remain inside task scope?
RD-PERMISSION	Were only granted permissions used?
RD-DATA	Were data boundaries respected?
RD-SECURITY	Did the output create security risk?
RD-LEGAL	Does the task require legal or jurisdictional handoff?
RD-SEMANTIC	Is the meaning correct and non-contradictory?
RD-ARCHITECTURE	Does the output fit the corpus architecture?
RD-TEST	Did tests/build/schema validation pass?
RD-WITNESS	Are required witness records present?
RD-ROLLBACK	Is rollback or correction path sufficient?
RD-MEMORY	Is memory promotion appropriate?
RD-RELEASE	Is public/release surface safe?
RD-CLOUD	Was cloud data exposure bounded?
RD-CORE	Does it touch identity, continuity, permission, or memory core?
RD-REDLINE	Does it approach prohibited action?

8.1 Minimum review by risk class

Risk	Required dimensions
R0	optional summary
R1	RD-SCOPE, RD-SEMANTIC
R2	RD-SCOPE, RD-PERMISSION, RD-TEST, RD-ROLLBACK
R3	R2 + RD-WITNESS, RD-RELEASE if public
R4	R3 + RD-MEMORY, RD-CORE, human gate
R5	R4 + RD-LEGAL, RD-SECURITY, preserve-before-repair
RX	RD-REDLINE, deny/quarantine

9. Consensus strength model

Consensus strength IDs use prefix CS-.*.

Level	Meaning	Use
CS-0	no consensus	hold/review
CS-1	weak agreement	low-risk support only
CS-2	moderate agreement	supports review, not final authority
CS-3	strong independent agreement	supports c decision
CS-4	strong agreement + tests + witness	supports integration where gates pass
CS-X	false or unsafe consensus	quarantine / review

9.1 Consensus weighting factors

Consensus weight increases when agents differ by:

- role;
- provider;
- runtime;
- input context;
- method;
- evidence source;
- test type;
- failure mode sensitivity.

Consensus weight decreases when agents share:

- provider;
- prompt;
- stale context;
- source document;

- model family;
- hidden assumption;
- same erroneous test;
- same incomplete task scope.

9.2 Consensus limitation rule

Even CS-4 consensus does not bypass:

- task contract;
- permission grant;
- sandbox/worktree profile;
- witness requirements;
- memory gate;
- c gate;
- human gate where required;
- legal/security review where required.

10. Same-source risk model

Same-source risk IDs use prefix SSR- *.

Level	Meaning	Default handling
SSR-0	independent sources and methods	normal weight
SSR-1	minor overlap	note limitation
SSR-2	shared input or prompt frame	reduce confidence
SSR-3	same provider/model family or same stale repo state	require independent check
SSR-4	circular review / one agent laundering another	hold/quarantine
SSR-X	fabricated or unsafe consensus	reject/quarantine

10.1 Same-source disclosure

A quorum record MUST disclose same-source risk.

10.2 Same-source mitigation

Mitigations include:

- local checker;
- independent source lookup inside authorized corpus;
- rerun from clean worktree;
- human review;

- different prompt/context;
- test fixture;
- witness chain check;
- delayed decision.

11. Disagreement taxonomy

Disagreement IDs use prefix DQ - *.

ID	Disagreement type	Meaning	Default action
DQ-NONE	no material disagreement	proceed to gate	
DQ-FACT	factual disagreement	verify source	
DQ-SCOPE	scope disagreement	hold / audit	
DQ-PERMISSION	permission disagreement	permission review	
DQ-TEST	test disagreement	rerun in clean sandbox	
DQ-SEMANTIC	meaning/architecture disagreement	c review	
DQ-RISK	risk classification disagreement	escalate review	
DQ-LEGAL	legal/jurisdictional concern	human/legal review	
DQ-MEMORY	memory promotion disagreement	memory gate review	
DQ-CORE	identity/continuity/privilege concern	human + c gate	
DQ-INCIDENT	incident interpretation disagreement	preserve + incident review	
DQ-REDLINE	prohibited action concern	deny/quarantine	
DQ-UNKNOWN	unclear disagreement	hold	

11.1 Disagreement rule

Disagreement is not failure.

Unclassified disagreement is failure.

11.2 Majority vote limitation

Majority vote MUST NOT override high-risk minority objection involving:

- red-line behavior;
- secrets;
- private/sealed/legal material;
- memory/core change;
- release/publication;
- incident evidence;

- external action.

12. Review decision vocabulary

Review decisions use prefix QRD-.*.

Decision	Meaning
QRD-ACCEPT	accept output as proposed
QRD-ACCEPT-WITH-LIMITS	accept only bounded subset
QRD-REVISE	return for revision
QRD-REJECT	reject output
QRD-HOLD	pause pending clarification
QRD-QUARANTINE	isolate output/agent/task
QRD-ROLLBACK	rollback applied or recommended
QRD-REHANDSHAKE	agent requires re-handshake
QRD-RECONTRACT	task requires new contract
QRD-C-GATE	send to c gate
QRD-HUMAN-GATE	human anchor required
QRD-LEGAL-REVIEW	legal/counsel route required
QRD-INCIDENT-REVIEW	incident/security route required
QRD-DENY-REDLINE	deny due to red-line

13. Review lifecycle

13.1 Standard lifecycle

```
artifact produced
-> review request
-> reviewer assignment
-> source/witness/diff/test refs loaded
-> review dimensions checked
-> same-source risk classified
-> disagreement classified
-> consensus strength assigned
-> review decision issued
-> witness event recorded
-> c gate / human gate / rollback / quarantine / integration
```

13.2 Pre-review checklist

Before review:

- task contract exists;
- output/diff/artifact exists;
- permission grant exists;
- sandbox/worktree refs exist;
- tests/validation available where applicable;

- witness refs available where required;
- denied path report available;
- cloud data classification available;
- rollback plan available for material changes.

13.3 Post-review checklist

After review:

- decision recorded;
- limitations recorded;
- disagreement recorded;
- consensus strength recorded;
- next gate identified;
- witness event emitted;
- memory gate recommendation separated from integration recommendation.

14. Quorum record object

Canonical object:

CLI_AGENT_QUORUM_RECORD

14.1 YAML shape

```
cli_agent_quorum_record:
  schema_version: cli-agent-quorum-review-0.1
  quorum_id: string
  created_at: string
  updated_at: string | null
  governing_entity_id: string
  governing_entity_name: string | null

task:
  task_id: string
  contract_id: string
  risk_class: R0 | R1 | R2 | R3 | R4 | R5 | RX
  objective_summary: string

quorum_profile:
  pattern: QP-1 | QP-2 | QP-3 | QP-4 | QP-5 | QP-6 | custom
  roles_required:
    - QROLE-EXECUTOR
    - QROLE-TESTER
    - QROLE-SEMANTIC
    - QROLE-AUDITOR
    - QROLE-C-GATE
  executor_reviewer_separation_required: true
  human_gate_required: boolean

participants:
  - participant_id: string
    agent_id: string | null
    agent_name: string | null
    provider: local | openai | google | anthropic | other | unknown | null
    runtime: local_cli | cloud_cli | api_agent | container_agent | hybrid | unknown | null
    role: QROLE-EXECUTOR | QROLE-TESTER | QROLE-SEMANTIC | QROLE-AUDITOR | QROLE-
ARCHIVIST | QROLE-SENTINEL | QROLE-JUDGE-ASSISTANT | QROLE-C-GATE | QROLE-HUMAN-GATE
    output_ref: string | null
    witness_ref: string | null

evidence_refs:
  diff_hash: string | null
  test_report_hash: string | null
  artifact_manifest_hash: string | null
  sandbox_run_ref: string | null
  permission_grant_ref: string | null
  witness_chain_ref: string | null
  rollback_plan_ref: string | null

review_summary:
  review_dimensions:
    - RD-SCOPE
    - RD-PERMISSION
    - RD-TEST
    - RD-SEMANTIC
    - RD-WITNESS
  same_source_risk: SSR-0 | SSR-1 | SSR-2 | SSR-3 | SSR-4 | SSR-X
  consensus_strength: CS-0 | CS-1 | CS-2 | CS-3 | CS-4 | CS-X
  disagreement_type: DQ-NONE | DQ-FACT | DQ-SCOPE | DQ-PERMISSION | DQ-TEST | DQ-SEMANTIC
| DQ-RISK | DQ-LEGAL | DQ-MEMORY | DQ-CORE | DQ-INCIDENT | DQ-REDLINE | DQ-UNKNOWN
  decision: QRD-ACCEPT | QRD-ACCEPT-WITH-LIMITS | QRD-REVISE | QRD-REJECT | QRD-HOLD |
QRD-QUARANTINE | QRD-ROLLBACK | QRD-REHANDSHAKE | QRD-RECONTRACT | QRD-C-GATE | QRD-HUMAN-
GATE | QRD-LEGAL-REVIEW | QRD-INCIDENT-REVIEW | QRD-DENY-REDLINE
  reason_code: string

authority:
  c_gate_required: true
  c_gate_ref: string | null
```

```
    human_gate_required: boolean
    human_gate_ref: string | null
    legal_review_required: boolean
    legal_review_ref: string | null

    witness:
      witness_required: boolean
      witness_event_ref: string | null
      append_only_required: true
      hash_required: boolean

    notes:
      summary: string | null
      limitations:
        - string
      assumptions:
        - string
      unresolved:
        - string
```

15. Review record object

Canonical object:

```
CLI_AGENT_REVIEW_RECORD
```

15.1 YAML shape

```
cli_agent_review_record:
  schema_version: cli-agent-quorum-review-0.1
  review_id: string
  created_at: string
  governing_entity_id: string
  reviewer:
    reviewer_type: agent | c | human_anchor | auditor | legal | security
    reviewer_id: string
    reviewer_role: QROLE-TESTER | QROLE-SEMANTIC | QROLE-AUDITOR | QROLE-ARCHIVIST | QROLE-
JUDGE-ASSISTANT | QROLE-C-GATE | QROLE-HUMAN-GATE

  subject:
    task_id: string
    contract_id: string
    agent_output_ref: string | null
    diff_hash: string | null
    artifact_hash: string | null
    sandbox_run_ref: string | null
    memory_proposal_ref: string | null
    release_package_ref: string | null
    incident_ref: string | null

  checks:
    dimensions_checked:
      - RD-SCOPE
      - RD-PERMISSION
      - RD-DATA
      - RD-SECURITY
      - RD-SEMANTIC
      - RD-TEST
      - RD-WITNESS
    passed: boolean
    uncertainty: none | low | medium | high | unknown
    risk_class_after_review: R0 | R1 | R2 | R3 | R4 | R5 | RX

  findings:
    accepted_points:
      - string
    rejected_points:
      - string
    concerns:
      - string
    scope_violations:
      - string
    redline_flags:
      - string

  decision:
    decision: QRD-ACCEPT | QRD-ACCEPT-WITH-LIMITS | QRD-REVISE | QRD-REJECT | QRD-HOLD |
QRD-QUARANTINE | QRD-ROLLBACK | QRD-REHANDSHAKE | QRD-RECONTRACT | QRD-C-GATE | QRD-HUMAN-
GATE | QRD-LEGAL-REVIEW | QRD-INCIDENT-REVIEW | QRD-DENY-REDLINE
    reason_code: string
    next_action: none | c_gate | human_gate | legal_review | incident_review | rollback |
quarantine | revise | revalidate

  witness:
    witness_required: boolean
    witness_event_ref: string | null
    append_only_required: true
```

16. Disagreement record object

Canonical object:

CLI_AGENT_DISAGREEMENT_RECORD

16.1 YAML shape

```
cli_agent_disagreement_record:
  schema_version: cli-agent-quorum-review-0.1
  disagreement_id: string
  created_at: string
  governing_entity_id: string
  quorum_id: string | null
  task_id: string
  contract_id: string

  disagreement:
    type: DQ-FACT | DQ-SCOPE | DQ-PERMISSION | DQ-TEST | DQ-SEMANTIC | DQ-RISK | DQ-LEGAL |
    DQ-MEMORY | DQ-CORE | DQ-INCIDENT | DQ-REDLINE | DQ-UNKNOWN
    severity: low | medium | high | critical
    summary: string
    affected_surfaces:
      - scope
      - permission
      - sandbox
      - memory
      - release
      - incident
      - core
      - cloud_data
      - witness

  positions:
    - participant_id: string
      agent_id: string | null
      position_summary: string
      evidence_ref: string | null
      confidence: low | medium | high | unknown

  handling:
    default_action: hold | revalidate | rerun_tests | c_review | human_review | legal_
    review | incident_review | quarantine | deny
    majority_vote_allowed: boolean
    minority_veto_reason: string | null

  outcome:
    decision: unresolved | resolved | escalated | quarantined | rejected | accepted_with_
    limits
    resolution_ref: string | null

  witness:
    witness_required: true
    witness_event_ref: string | null
```

17. Consensus limit record object

Canonical object:

CLI_AGENT_CONSENSUS_LIMIT_RECORD

17.1 YAML shape

```
cli_agent_consensus_limit_record:
  schema_version: cli-agent-quorum-review-0.1
  consensus_limit_id: string
  created_at: string
  governing_entity_id: string
  quorum_id: string
  consensus_strength: CS-0 | CS-1 | CS-2 | CS-3 | CS-4 | CS-X
  same_source_risk: SSR-0 | SSR-1 | SSR-2 | SSR-3 | SSR-4 | SSR-X

  limitation_reasons:
    - shared_provider
    - shared_prompt
    - shared_context
    - stale_repo_state
    - same_test_fixture
    - missing_witness
    - no_local_checker
    - no_human_gate
    - cloud_only_review
    - high_uncertainty

  effect:
    may_support_integration: boolean
    may_support_memory_promotion: boolean
    may_support_release: boolean
    requires_additional_review: boolean
    required_next_review: none | local_checker | human_gate | c_gate | legal_review |
    incident_review | revalidation

  witness:
    witness_required: boolean
    witness_event_ref: string | null
```

18. Review event families

Event families use prefix:

```
cli_agent.quorum.*
cli_agent.review.*
cli_agent.disagreement.*
```

18.1 Quorum event families

Event family	Meaning
cli_agent.quorum.created	quorum record created
cli_agent.quorum.started	quorum review started
cli_agent.quorum.participant_added	participant added
cli_agent.quorum.participant_removed	participant removed
cli_agent.quorum.completed	quorum review completed
cli_agent.quorum.split	material disagreement remains
cli_agent.quorum.consensus_limited	consensus limitation recorded
cli_agent.quorum.escalated	escalated to c, human, legal, or incident review
cli_agent.quorum.invalidated	quorum invalidated due to scope/source/witness issue

18.2 Review event families

Event family	Meaning
<code>cli_agent.review.requested</code>	review requested
<code>cli_agent.review.started</code>	review started
<code>cli_agent.review.completed</code>	review completed
<code>cli_agent.review.accepted</code>	output accepted
<code>cli_agent.review.accepted_with_limits</code>	limited acceptance
<code>cli_agent.review.revise_required</code>	revision required
<code>cli_agent.review.rejected</code>	output rejected
<code>cli_agent.review.quarantined</code>	output quarantined
<code>cli_agent.review.rollback_recommended</code>	rollback recommended
<code>cli_agent.review.redline_detected</code>	red-line detected

18.3 Disagreement event families

Event family	Meaning
<code>cli_agent.disagreement.detected</code>	disagreement detected
<code>cli_agent.disagreement.classified</code>	disagreement classified
<code>cli_agent.disagreement.resolved</code>	disagreement resolved
<code>cli_agent.disagreement.escalated</code>	disagreement escalated
<code>cli_agent.disagreement.minority_veto</code>	high-risk minority concern blocks majority acceptance
<code>cli_agent.disagreement.unresolved_hold</code>	unresolved disagreement causes hold

19. Review handling by surface

19.1 Code/schema surface

Required checks:

- scope;
- diff;
- tests;
- rollback;
- no denied paths;
- no unapproved dependency changes.

19.2 Protocol/document surface

Required checks:

- terminology;
- contradiction;
- duplication;

- anti-echo;
- corpus bridge;
- non-goals;
- red-line boundaries;
- machine-facing metadata if relevant.

19.3 Release/public surface

Required checks:

- build;
- links;
- checksums;
- metadata;
- discoverability;
- protected branch rules;
- human gate.

19.4 Memory/immunity surface

Required checks:

- memory class;
- source refs;
- witness refs;
- poisoning risk;
- no retaliation;
- rollback/disable path;
- c gate;
- human gate for high-risk.

19.5 Incident surface

Required checks:

- preserve-before-repair;
- minimal evidence;
- legal/security boundary;
- no live counter-operation;
- incident witness;
- human gate where sensitive.

19.6 Core authority surface

Required checks:

- direct mutation prohibited;
- c gate;
- human gate;
- witness;
- delayed review where possible;
- rollback/correction plan.

20. Human and c gate rules

20.1 c gate

The c gate decides whether reviewed output becomes:

- accepted task output;
- operational note;
- memory candidate;
- reviewed memory;
- experience artifact;
- defensive immunity update;
- release/public surface;
- rejected output;
- quarantined output;
- rollback trigger.

20.2 Human gate

Human gate is required for:

- R4/R5 tasks;
- core-memory proposals;
- identity/continuity/permission changes;
- high-risk defensive immunity updates;
- release/publication if irreversible or reputationally significant;
- legal-sensitive or incident-sensitive outputs;
- no-rollback actions;
- red-line-adjacent cases.

20.3 Gate order

Default order:

```
agent review
-> quorum record
-> c gate
-> human gate where required
-> witness
-> integration / memory gate / release / rollback
```

21. Red-line review handling

21.1 Red-line conditions

A review MUST classify as red-line if output contains or recommends:

- hack-back;
- live external counter-operation;
- malware behavior;
- credential theft;
- covert persistence;
- evasion;
- unauthorized scanning;
- direct memory write;
- direct core mutation;
- self-approval;
- witness tampering;
- secret export;
- cloud leakage of prohibited material;
- public release without gate.

21.2 Required red-line response

```
deny
quarantine
freeze affected path
revoke or narrow permission if needed
record witness
human review
no automatic re-entry
```

21.3 No majority override

No majority of agents may override a red-line classification.

22. Validation workflow

```

parse quorum/review record
-> validate participants and roles
-> check executor/reviewer separation
-> load task contract and permission refs
-> load diff/test/witness refs
-> classify review dimensions
-> classify same-source risk
-> classify disagreement
-> assign consensus strength
-> check human/`c` gate requirements
-> issue review decision
-> witness review outcome
-> route to integration, memory gate, rollback, quarantine, or escalation

```

23. Failure mapping

Failure	Required default
executor self-reviews material output	hold or quarantine
no reviewer for R2+ material change	hold
no local/test validation for code/schema task	hold or revise
same-source risk high and unacknowledged	hold
disagreement unclassified	hold
red-line minority concern ignored	quarantine
missing witness for review-required transition	hold
quorum tries to bypass c gate	deny
quorum tries to bypass human gate	deny
cloud-only review of sensitive material	hold / human_review
review output contains prohibited material	quarantine
consensus used as direct memory write	reject / quarantine

24. Conformance levels

Level	Meaning
QRP-0	no quorum/review discipline
QRP-1	manual review only
QRP-2	executor/reviewer separation for material changes
QRP-3	structured review records and disagreement classification
QRP-4	quorum records, same-source risk, witness-linked review decisions
QRP-5	high assurance: multi-role quorum, local checks, memory/release/incident gates, conformance tests
QRP-X	non-conformant / self-approval / red-line consensus laundering

25. Mandatory conformance gates

Gate	Name	Blocking failure
G0	Role separation	executor can final-approve own work
G1	Review record	material change lacks review record
G2	Evidence refs	review lacks diff/test/source/witness refs
G3	Disagreement classification	material disagreement unclassified
G4	Same-source risk	consensus treated as independent without basis
G5	c gate	quorum bypasses c decision
G6	Human gate	high-risk task bypasses human gate
G7	Memory gate	quorum output enters memory directly
G8	Red-line handling	prohibited action accepted by majority
G9	Witness	review-required transition unwitnessed
G10	Quorum boundary	agent group creates tasks/permissions/tools without governance

26. Red-line failures

A system **MUST** be classified as QRP-X if:

1. an executor final-approves its own material work;
2. a quorum grants itself new permissions;
3. a quorum bypasses c gate;
4. a quorum bypasses human gate for R4/R5/core/incident/release tasks;
5. consensus is used to authorize hack-back or live counter-operation;
6. consensus is used to promote malware-like or credential-theft behavior;
7. same-source consensus is misrepresented as independent high confidence;
8. red-line minority objection is ignored;
9. quorum output writes memory directly;
10. quorum mutates identity/core directly;
11. review suppresses test failures;
12. review hides scope violations.

27. Examples

27.1 Codex + Gemini + local checker quorum

```
cli_agent_quorum_record:
  schema_version: cli-agent-quorum-review-0.1
  quorum_id: qr-20260516-codex-gemini-local-001
  created_at: "2026-05-16T23:30:00Z"
  updated_at: null
  governing_entity_id: ester
  governing_entity_name: Ester

task:
  task_id: task-cli-schema-patch-001
  contract_id: catc-cli-schema-patch-001
  risk_class: R2
  objective_summary: Patch Markdown table formatting and validate schema references.

quorum_profile:
  pattern: QP-3
  roles_required:
    - QROLE-EXECUTOR
    - QROLE-SEMANTIC
    - QROLE-TESTER
    - QROLE-C-GATE
  executor_reviewer_separation_required: true
  human_gate_required: false

participants:
  - participant_id: p1
    agent_id: codex-executor-01
    agent_name: Codex Executor
    provider: openai
    runtime: cloud_cli
    role: QROLE-EXECUTOR
    output_ref: diff-docs-format-001
    witness_ref: we-codex-run-001
  - participant_id: p2
    agent_id: gemini-reader-01
    agent_name: Gemini Reader
    provider: google
    runtime: cloud_cli
    role: QROLE-SEMANTIC
    output_ref: semantic-review-001
    witness_ref: we-gemini-review-001
  - participant_id: p3
    agent_id: local-checker-01
    agent_name: Local Checker
    provider: local
    runtime: local_cli
    role: QROLE-TESTER
    output_ref: test-report-001
    witness_ref: we-local-test-001

evidence_refs:
  diff_hash: diff-hash-001
  test_report_hash: test-report-hash-001
  artifact_manifest_hash: null
  sandbox_run_ref: run-cli-schema-patch-001
  permission_grant_ref: grant-cli-schema-patch-001
  witness_chain_ref: chain-task-cli-schema-patch-001
  rollback_plan_ref: rb-cli-schema-patch-001

review_summary:
  review_dimensions:
    - RD-SCOPE
```

```
- RD-PERMISSION
- RD-TEST
- RD-SEMANTIC
- RD-WITNESS
- RD-ROLLBACK
same_source_risk: SSR-1
consensus_strength: CS-3
disagreement_type: DQ-NONE
decision: QRD-C-GATE
reason_code: review_passed_pending_c_gate

authority:
  c_gate_required: true
  c_gate_ref: null
  human_gate_required: false
  human_gate_ref: null
  legal_review_required: false
  legal_review_ref: null

witness:
  witness_required: true
  witness_event_ref: we-quorum-001
  append_only_required: true
  hash_required: true

notes:
  summary: Executor patch, semantic review, and local tests aligned. c gate still
required.
  limitations:
    - Codex and Gemini are both cloud-origin; no private material was included.
  assumptions:
    - Worktree was clean before execution.
  unresolved: []
```


27.2 High-risk disagreement blocks majority

```
cli_agent_disagreement_record:
  schema_version: cli-agent-quorum-review-0.1
  disagreement_id: dq-20260516-redline-001
  created_at: "2026-05-16T23:45:00Z"
  governing_entity_id: liya
  quorum_id: qr-incident-review-001
  task_id: task-defensive-emulation-001
  contract_id: catc-defensive-emulation-001

disagreement:
  type: DQ-REDLINE
  severity: critical
  summary: One reviewer flags proposed immunity update as live counter-operation risk.
  affected_surfaces:
    - incident
    - memory
    - core
    - witness

positions:
  - participant_id: p1
    agent_id: local-sentinel-01
    position_summary: Proposed update should block and quarantine matching channel.
    evidence_ref: immunity-proposal-001
    confidence: medium
  - participant_id: p2
    agent_id: auditor-01
    position_summary: Proposed wording could authorize active response against external
source.
    evidence_ref: audit-risk-001
    confidence: high

handling:
  default_action: quarantine
  majority_vote_allowed: false
  minority_veto_reason: Red-line concern: possible live counter-operation.

outcome:
  decision: quarantined
  resolution_ref: null

witness:
  witness_required: true
  witness_event_ref: we-disagreement-redline-001
```

Required outcome:

```
quarantine proposal
revise to defensive-only wording
human + c review
no live counter-operation
```

27.3 Invalid self-approval pattern

```
cli_agent_review_record:
  schema_version: cli-agent-quorum-review-0.1
  review_id: review-invalid-self-approval-001
  created_at: "2026-05-17T00:00:00Z"
  governing_entity_id: ester
  reviewer:
    reviewer_type: agent
    reviewer_id: codex-executor-01
    reviewer_role: QROLE-EXECUTOR

  subject:
    task_id: task-release-patch-001
    contract_id: catc-release-patch-001
    agent_output_ref: diff-release-patch-001
    diff_hash: diff-release-patch-hash-001
    artifact_hash: null
    sandbox_run_ref: run-release-patch-001
    memory_proposal_ref: null
    release_package_ref: release-package-001
    incident_ref: null

  checks:
    dimensions_checked:
      - RD-SCOPE
      - RD-RELEASE
    passed: true
    uncertainty: low
    risk_class_after_review: R3

  findings:
    accepted_points:
      - Executor claims release patch is ready.
    rejected_points: []
    concerns:
      - Same agent produced and approved the patch.
    scope_violations: []
    redline_flags:
      - self_approval

  decision:
    decision: QRD-QUARANTINE
    reason_code: self_approval_attempt
    next_action: quarantine

  witness:
    witness_required: true
    witness_event_ref: we-self-approval-001
    append_only_required: true
```

Required result:

```
quarantine review
require independent reviewer
no release integration
```

28. Standard reason codes

28.1 Acceptance codes

- scope_valid
- tests_passed
- semantic_consistent
- witness_present
- rollback_present
- independent_review_passed
- consensus_strong_bounded

28.2 Revision codes

- minor_scope_gap
- tests_missing
- semantic_ambiguity
- terminology_drift
- rollback_incomplete
- witness_incomplete
- same_source_risk

28.3 Rejection/quarantine codes

- scope_violation
- permission_violation
- denied_path_attempt
- secret_exposure
- private_data_exposure
- redline_detected
- self_approval_attempt
- consensus_laundering
- core_mutation_risk
- memory_poisoning_risk
- live_counteroperation_risk

29. Implementation notes

29.1 Reviewer prompt isolation

Reviewers SHOULD receive task contract, diff, tests, and relevant context, not an unlimited project dump.

29.2 Reviewer independence

Independence is improved by different provider/runtime/method, but role separation matters more than vendor diversity alone.

29.3 Local checker value

A local checker is valuable because it verifies concrete state: files, tests, schemas, hashes, builds. It reduces cloud-only consensus risk.

29.4 Human-readable review card

A review should produce a short card:

- Task:
- Executor:
- Reviewer(s):
- Tests:

Scope:
Risks:
Disagreement:
Decision:
Next gate:

29.5 Do not overvote safety

Safety objections are not popularity contests. One serious red-line objection is enough to hold or quarantine.

29.6 Quorum expiry

Quorum decisions should expire when repository state, task scope, model/tool versions, or risk classification changes materially.

30. Open issues

ID	Issue	Required action
0I-001	JSON Schema extraction	Extract quorum/review/disagreement objects to <code>.schema.json</code> .
0I-002	Standard prompt cards	Define bounded prompts for executor, tester, semantic reviewer, auditor.
0I-003	Independence scoring	Define numeric scoring for same-source risk.
0I-004	Local checker profile	Create detailed local checker / tester companion.
0I-005	Human gate UI	Define human approval display and stop-rule language.
0I-006	Memory gate binding	Align quorum output with Memory Gate Profile.
0I-007	Release quorum binding	Align with future Release/Public Surface Profile.
0I-008	Incident quorum binding	Align with future Incident Response Profile.
0I-009	Red-line veto handling	Define exact minority-veto procedure.
0I-010	Repo placement	Decide final GitHub path and package index integration.

31. Closing rule

Quorum is useful because no single worker sees the whole system.

Quorum is dangerous when the worker group begins to believe it is the system.

Final rule:

Many agents may review.
Only c integrates.
Human anchors irreversible consequence.